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Printed Pages—7 **EE302**

(Following Paper ID and Roll No. to be filled in your Answer Book)

PAPER ID : 1253 **Roll No.** | | | | | | | | | | |

B.Tech.

(SEM. III) ODD SEMESTER THEORY EXAMINATION 2013-14

BASICS OF SIGNALS AND SYSTEMS

Time : 3 Hours *Total Marks : 100*

Note :– Attempt questions from all Sections as per directions.

SECTION–A

1. Attempt **all** parts : **(2×10=20)**

   (a) What is the difference between continuous-time and discrete-time signals ?

   (b) Define unit impulse function $\delta(t)$.

   (c) Explain the analogy between mechanical and electrical systems.

   (d) What are the dirichlet conditions for the existence of Fourier Series ?

   (e) Find the Fourier transform of Unit-Step function.

   (f) Prove the time-shift property for Laplace transform.

   (g) Explain the applications of Laplace transform.

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(h) Explain the terms 'state' and 'state variables' with examples.

(i) List the advantages of state space representation of Linear Systems.

(j) Explain initial and Final Value Theorem for z-transform.

SECTION–B

2. Attempt **three** parts : **(3×10=30)**

(a) Express the signal shown in Fig. 1 in terms of step signals.

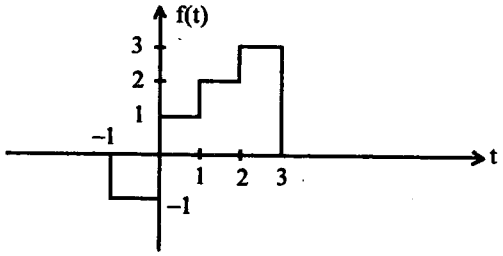


Fig. 1

(b) Draw the force-current analogy of the mechanical system shown in Fig. 2.

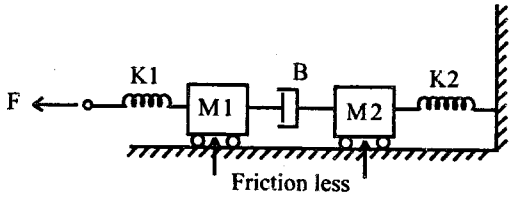


Fig. 2

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(c) Find the Fourier transform of signal shown in Fig. 3.

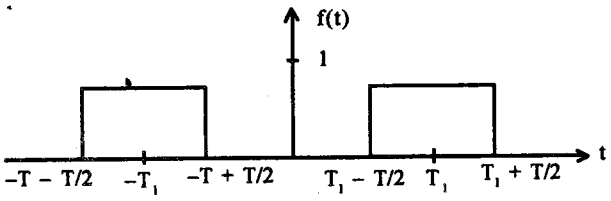


Fig. 3

(d) Find the inverse Laplace transform of following function :

$$F(s) = \frac{3s^2 + 8s + 6}{(s + 2)(s^2 + 2s + 1)}$$

(e) Define the state transition matrix and its properties.

SECTION–C

Note :– Attempt **all** questions from this Section. **(10×5=50)**

3. Attempt any **two** parts of the following :

(a) Distinguish between Periodic and Non-periodic Signals.

Find the time-period of the signal $x(t) = \cos\frac{\pi}{3}t + \sin\frac{\pi}{4}t$.

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(b) Synthesize the signal (Fig. 4) using basic signals.

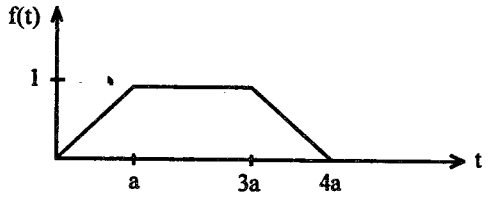


Fig. 4

(c) What is a LTI system ? Check whether the system $y(t) = x^2(t)$ is a LTI system.

4. Attempt any **one** part of the following :

(a) Determine the Fourier series for a square wave signal of unity magnitude using trigonometric series.

(b) Find the exponential Fourier series of the signal shown in Fig. 5 :

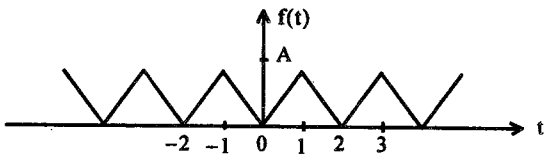


Fig. 5

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5. Attempt any **two** parts of the following :

(a) Find the Laplace transform of function $x(t) = Ae^{-a|t|}$.

(b) Find the inverse Laplace transform of the function :

$$F(s) = \frac{(s+1) + 3.e^{-4s}}{(s+2)(s+3)}$$

(i) ROC : $Re\{s\} > 3$

(ii) ROC : $Re\{s\} < 2$.

(c) Consider the circuit shown in Fig. 6. Initially the switch is in position 1. At t = o, the switch is moved to position 2. Find the expression for the current in the inductor L.

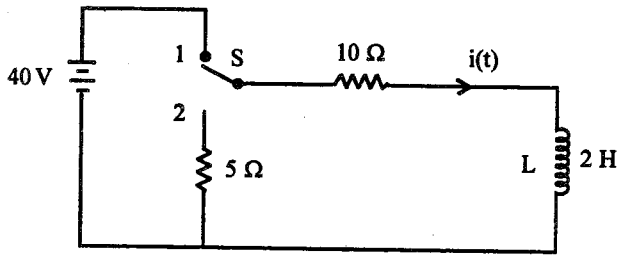


Fig. 6

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6. Attempt any **one** part of the following :

(a) State space representation of a system is given by :

$$\dot{X}(t) = \begin{bmatrix} 0 & 1 \\ -2 & -3 \end{bmatrix} x(t) + \begin{bmatrix} 0 \\ 1 \end{bmatrix} u(t) \, , \; y(t) = [1 \quad 0] x(t)$$

where u(t) is the unit step input. All the initial conditions are zero. Find the time-response of the system.

(b) Develop the state model for the circuit shown in Fig. 7 :

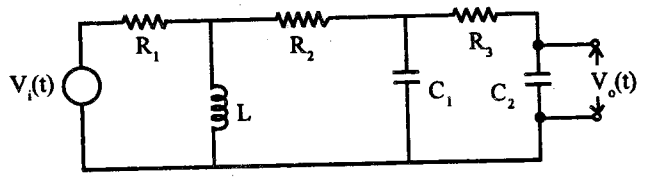


Fig. 7

7. Attempt any **two** parts of the following :

(a) Find the z-transform of $x[n] = \sin w_0 n \, u[n]$.

(b) Find the inverse z-transform of function :

$$X(z) = \frac{7z - 23}{(z-3)(z-4)} \quad |z| > 4$$

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(c) Find the response of the following difference equation for step input. Assume zero initial conditions :

$$y[n] - \frac{5}{6}y[n-1] + \frac{1}{6}y[n-2] = x[n] - \frac{1}{2}x[n-1].$$

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